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PROJECT REPORT

The use of ML prediction and classification in judicial contexts

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1 Introduction

1.1 Context

My research project focuses on the application of predictive machine learning (ML) in judicial contexts. To develop this report, I primarily relied on the work of Masha Medvedeva - a researcher specializing in automation within the legal domain - detailed in her article *Using machine learning to predict decisions of the European Court of Human Right* (2019). [1]

Medvedeva and her colleagues conducted a study to investigate the potential of supervised ML in predicting legal outcomes, particularly violations of the European Convention on Human Rights (ECHR). Using Support Vector Machines (SVMs) and textual features, their research aimed to classify cases based on their outcomes, demonstrating both the promise and challenges of such approaches in the legal field.

One of the primary objectives of my project stems from a critical analysis of an article by Aletras et al. [2], which aimed to detect whether there was a violation of a specific article of the European Convention on Human Rights. The intended purpose of their approach was to help alleviate the judges' workload by identifying whether a case fell within the Court's competence. While their reported performance reached an accuracy of 0.8, which is relatively high, the remaining 20% margin of error raises critical concerns. Specifically, a major limitation of their methodology lies in their evaluation process : 70% of the dataset included cases where the decision was explicitly stated, while the remaining 30% excluded this part of the judgment text. However, simply removing the section containing the decision does not entirely eliminate bias, as the judgment's phrasing is written after the decision has already been made.

An alternative approach, and one of the key contributions of this project, is to utilize the procedural and factual descriptions available prior to the judgment. By leveraging these pre-decision elements, the prediction task becomes more robust and less dependent on the post-judgment text.

1.2 Goals

This consideration highlights my first objective : reviewing the state of the art which played a critical role in guiding my project. Importantly, the project required careful examination of the term "prediction," as its meaning carries significantly different connotations in the domains of machine learning and law. This dual perspective shaped my methodological choices, particularly my decision to build upon the work of Medvedeva. Her research provided a well-structured framework and nomenclature, which was instrumental in navigating this domain as a novice. Her contributions enabled me to align my project with established methodologies while minimizing the risk of errors. Indeed, as the first student to join the ARIA from the English department, I was confronted to several initial challenges regarding the assimilation of an entirely new field to work on. My knowledge of coding was rudimentary at best and managing to not only recreate an existing code, but also to use it as the foundation for something new represented a sizeable effort for a student coming from a (up to this point) purely literary background.

The second major goal of this project was to replicate Medvedeva's work. Reproducing her methodology served as a crucial foundation, ensuring that I could validate and test my understanding of the approaches she developed and refine my ability to identify different AI architectures, training processes, and validation strategies.

Finally, the third goal was to extend Medvedeva's work by further analyzing results at a more granular level, specifically breaking down performance metrics by year. This additional effort represents an original contribution to the field, showcasing how my project moves beyond simple replication to generate new insights.

2 State of the art : ML prediction in the legal domain

2.1 Theoretical backround

This subsection explores the theoretical foundations of prediction in ML, with a focus on recent developments in classifying court outcomes. The use of ML in this context has grown remarkably over the past six years (2015–2021), with numerous studies aiming to predict decisions of courts in different parts of the world, each with their own specific legal system and laws. However, some researchers remain wary of the fact that some of these studies may not genuinely achieve what they claim – that is, “predicting” court decisions.

Indeed, the word “prediction” possesses a different definition in the judicial field than it does in the field of machine learning. According to Medvedeva [1], many studies claiming to predict court decisions perform one of three distinct tasks, which are often conflated under the umbrella term "legal decision prediction."

2.1.1 Three Core Tasks in Legal Decision Prediction

Legal decision prediction, as a term, encompasses different tasks that fall within the broader realm of classification tasks in ML. These include :

- **1. Outcome Identification :** This task involves extracting the verdict from the text of a published court judgment. It focuses on analyzing the textual content of legal documents to identify whether the defendant won or lost, using techniques from natural language processing (NLP).
- **2. Outcome-Based Judgment Categorization :** This refers to classifying legal documents based on their outcomes, with the aim of studying contentious cases of alimony in divorce settlements for example. Researchers extract descriptive features of cases (e.g., socioeconomic factors, case history) to estimate critical metrics such as monetary awards or compensation amounts. The primary goal here is to identify the variables that may influence a judge’s decision and determine their relative importance.
- **3. Outcome Forecasting :** Unlike the previous tasks, outcome forecasting aims to predict court decisions before they are made. This task is more ambitious, as it involves forecasting judicial outcomes for ongoing or future cases, relying on insights from data collected from past decisions.

The type of task that an ML system can perform largely depends on the nature of the data available. As Medvedeva (2020a) [3] highlighted, distinctions between these tasks are not always clearly made in the legal field, leading to ambiguities in understanding their purpose.

2.1.2 The Role of Supervised Learning in Legal Decision Prediction

All three tasks share a common methodological framework : **supervised learning**. In supervised learning, the system is trained on labeled input data, such as facts extracted from criminal cases paired with their outcomes (e.g., "prosecution won" or "defendant won"). During training, the ML model learns patterns that link input data to corresponding labels. Once trained, the model is tested on data which has not been used during the training phase, such as new criminal cases, to predict the label for each document. While identifying these tasks as "predicting court decisions" is methodologically accurate within the ML and NLP domains, the term "prediction" can be misleading outside these fields. In the legal domain, "prediction" often implies forecasting decisions that judges have yet to make. However, most ML studies in this area focus on cases where decisions are already known and thus do not engage in true forecasting.

2.1.3 Ambiguity in Terminology and Implications

This ambiguity stems from the overlap in ML terminology and the expectations within the legal field. In NLP, "prediction" refers to generating outputs based on learned patterns from training data. Conversely, in the judicial field, the term suggests an ability to anticipate judicial decisions in advance. As a result, many systems claiming to predict court decisions do not attempt to forecast outcomes for

cases that are still undecided. This distinction is critical to understanding the capabilities and limitations of ML in the legal context and has significant implications for its application in practice.

2.2 Outcome identification

Outcome identification refers to the task of determining the verdict in legal cases by analyzing the text of published court judgments. In the context of the European Court of Human Rights (ECtHR), this task involves extracting textual information from relevant sections of the judgments that reference the final decision (e.g., phrases like "therefore, there is a violation of Article 3"). For a more in-depth explanation of the different components of ECtHR judgments, see the section on data.

Aletras et al. (2016) [2] pioneered this task by using a support vector machine (SVM) to predict court decisions based on textual features extracted from ECtHR judgments. Their model focused on identifying the court's outcome by analyzing key sections of the text without directly including the final verdict. They achieved an average accuracy of 79% across three articles of the European Convention on Human Rights (ECHR).

Paper	Court	Max. performance
Aletras et al. (2016)	ECtHR	79%
Liu and Chen (2017)	ECtHR	88%
Sulea et al. (2017a, b)	French Court of Cassation	99%
Virtucio et al. (2018)	Philippine Supreme Court	59%
Lage-Freitas et al. (2019)	Brazilian courts (appeal)	79% (F1)
Visentin et al. (2019)	ECtHR	79%
Bertalan and Ruiz (2020)	São Paolo Justice Court	98% (F1)
Quemy and Wrembel (2020)	ECtHR	96%

FIGURE 1 – Research that falls under the category of outcome identification[4]

Building on Aletras et al.'s work, Liu and Chen (2017) [5] and Visentin et al. (2019) [6] applied similar statistical techniques to the same dataset. Liu and Chen improved the task's accuracy to 88% by employing an SVM model with additional refinements. Meanwhile, Quemy and Wrembel (2020) expanded the dataset significantly and used neural network models to classify cases as either a violation or non-violation of any article of the ECHR. Although their task also fell under outcome identification, they included all sections of the judgment in their analysis. This comprehensive approach resulted in an impressive accuracy of 96% across a range of statistical methods.

While these studies perform tasks that can be classified as "prediction" in machine learning terms, they do not actually forecast decisions that have yet to be made. As a result, their practical utility in the judicial field remains quite limited.

2.3 Outcome-based judgement identification

Outcome-based judgment categorisation involves grouping court decisions based on their outcomes, such as whether they indicate a violation of a legal article. Unlike outcome identification, this task seeks to identify the predictors or key features that explain the categorisation. As argued by Medvedeva et al. (2020) [3], this task provides limited utility for already published judgments, as the outcomes are readily accessible. Instead, its value lies in uncovering the most informative features that drive classification. For this reason, it is crucial that the underlying machine learning (ML) models are interpretable and not 'black boxes' – which is also why I opted not to use LinkBERT for my project, as BERT (Bidirectional Encoder Representations from Transformers) models are considered to be 'black boxes'.

Medvedeva et al. (2020)[3] achieved an accuracy of up to 75% by categorising decisions of the European Court of Human Rights (ECtHR) using only the factual sections of the judgments. They employed

support vector machines (SVMs), a method that allows for the identification of top predictors—specific sequences of one or more words—that inform the classification. This interpretability is a key advantage of SVMs compared to more complex models like neural networks based on BERT.

Paper	Court	FI	Max. performance
Kowsrihawat et al. (2018)	Thai Supreme Court	×	67%
Chalkidis et al. (2019)	ECtHR	✓	82% (F1)
Kaufman et al. (2019)	SCOTUS	×	77%
Kaur and Bozic (2019)	ECtHR	×	82%
O'Sullivan and Beel (2019)	ECtHR	×	69%
Chalkidis et al. (2020)	ECtHR	×	83% (F1)
Condevaux (2020)	ECtHR	×	88%
Medvedeva et al. (2018, 2020a)	ECtHR	✓	75%
Salaün et al. (2020)	Québec Rental Tribunal	×	85%
Shaikh et al. (2020)	Dehli District Court	×	92%
Strickson and De La Iglesia (2020)	UK highest Court of Appeal	✓	69%
Sert et al. (2021)	Turkish Constitutional Court	×	98% (F1)
Malik et al. (2021)	Indian Supreme Court Court	×	77%
Medvedeva et al. (2021)	ECtHR	×	92% (F1)

FIGURE 2 – Research that falls under the category of outcome-based judgement identification[4]

Chalkidis et al. (2019) [7] also focused on ECtHR judgments used neural models, including a hierarchical attention network, to enhance categorisation accuracy. Their approach reached a performance of up to 82% and, like SVMs, allowed for the identification of critical words and facts. However, Chalkidis et al. (2020) later proposed the use of LEGAL-BERT, a neural architecture fine-tuned for legal data. While LEGAL-BERT demonstrated superior performance (F1-score of 83%), it sacrificed interpretability due to the complexity of the model. This limitation undermines the primary goal of this task, which is to identify clear predictors for categorisation.

Medvedeva et al. (2021) [4] replicated the experiments from Chalkidis et al. (2019, 2020) to compare SVMs, neural networks, and LEGAL-BERT for categorisation and forecasting tasks. For categorising ECtHR cases from 2019, their implementation of LEGAL-BERT achieved an F1-score of 92%. However, its performance varied significantly depending on the year, with a drop to 62% for 2020 cases, highlighting the sensitivity of these models to temporal shifts in data.

The trade-off between performance and interpretability remains a central challenge in outcome-based judgment categorisation. While neural models like LEGAL-BERT offer higher accuracy, SVMs provide insights into the features influencing categorisation, making them a valuable choice when interpretability is critical.

2.4 Outcome forecasting : JURI SAYS

Outcome forecasting refers to the prediction of court decisions before they are rendered, using historical data and ML techniques. Unlike outcome identification or categorisation, this task focuses on forecasting judgments of cases that are yet to be decided. This task proves to be particularly challenging in light of the inherent temporal and societal dynamics that shape legal decisions. Medvedeva and her team have worked on outcome forecasting by developing the JURI SAYS model, which I initially intended to analyze for this project. Unfortunately, while the model is accessible via a web-based platform, its code was not publicly available : this limitation led me to focus on a different approach. However, the significance of the JURI SAYS model warrants attention, which is why I chose to dedicate a portion of my report to providing a more detailed explanation of its workings.

JURI SAYS represents a notable achievement in forecasting the outcome of decisions of the European Court of Human Rights (ECtHR). The system is designed to predict the court's decisions using communicated cases, which are published long before the final rulings are made. Importantly, JURI SAYS explicitly avoids training its models on the final decisions, ensuring that its predictions are based solely on past cases. This methodology eliminates any bias that could arise from using information unavailable at the time of the decision-making process.

FIGURE 3 – JURI SAYS frontpage [3]

The system comprises three independent components : a database, an ML system, and a web-based platform. This modular design allows for flexibility and adaptability in refining each component without disrupting the others. The data pipeline involves monthly updates from HUDOC, the ECtHR's official database. New communicated cases are downloaded, and the ML model is retrained using all prior cases, excluding those from the most recent month. This ensures that the model remains up-to-date and capable of making predictions on the latest set of cases.

In addition to the communicated cases, JURI SAYS incorporates historical data from the last ten years, including non-communicated cases, to increase the dataset's size and diversity. The training data includes both the full-text judgments and associated metadata, such as parties involved, referenced articles, and case outcomes.

2.4.1 Machine Learning architecture

JURI SAYS employs a binary classification approach to predict whether a case will result in a violation or non-violation of human rights. Beyond predicting the case outcome, it performs a granular sentence-level probability analysis. Each sentence is evaluated for its likelihood of belonging to a case with a violation versus a non-violation. This probability is visualized in the system's output, with green highlighting for sentences aligning with the predicted decision and red for those that contradict it. These visual cues provide users with insights into the reasoning behind the model's predictions, enhancing interpretability.

FIGURE 4 – An example of a correctly predicted case by JURI SAYS with highlighted sentences. [3]

The sentence-level analysis not only improves prediction accuracy but also helps to understand the key textual elements that influence the model's decision. Such interpretability is critical in legal contexts, where the justification for a decision is as important as the decision itself.

2.4.2 Performance

The JURI SAYS forecasting model achieved an accuracy of 75% using SVMs. The model is retrained monthly to ensure that predictions reflected the latest available data. This monthly retraining aligns with the dynamic nature of ECtHR judgments, which can evolve with shifting societal contexts and legal interpretations.

JURI SAYS thus outperforms earlier approaches like those by Katz et al. (2017) [8], who employed random forests and reported an accuracy of 70%. JURI SAYS benefits from its more sophisticated architecture, larger dataset, and continuous model updates, resulting in significantly higher accuracy scores. According to Medvedeva, JURI SAYS consistently achieves robust results, as detailed in performance tables. However, like all forecasting models, its performance may vary depending on changes in legal interpretation, societal norms, or the availability of new evidence.

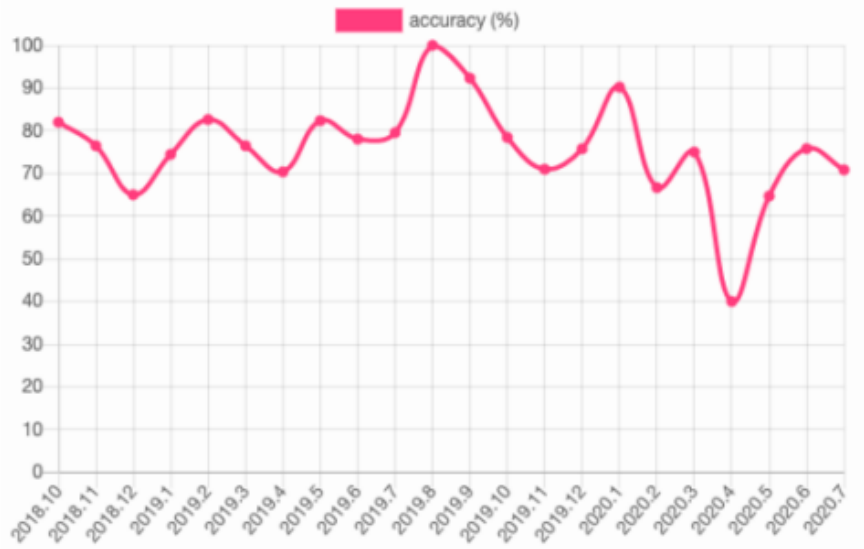

FIGURE 5 – Accuracy of JURI SAYS over the years 2018 - 2020 predicting future judgements. [3]

Paper	Court	Data	Max. performance
Sharma et al. (2015)	SCOTUS	Court of Appeal info	70%
Katz et al. (2017)	SCOTUS	Court of Appeal info	70%
Waltl et al. (2017)	German Court of Appeal (Tax Law)	Decision of the lower (fiscal) courts	57% (F1)
Medvedeva et al. (2020b)	ECtHR	Facts as communicated to the parties	75%
Medvedeva et al. (2021)	ECtHR	Facts as communicated to the parties	66% (F1)

FIGURE 6 – Research that falls under the category of outcome forecasting [4]

2.4.3 Challenges in outcome forecasting

Despite its advancements, outcome forecasting faces unique challenges that differ from other ML tasks in the legal domain. The temporal nature of the task means that the model must contend with evolving legal standards, societal changes, and shifts in judicial interpretation. Reliance on historical data can disadvantage forecasting systems compared to those that classify judgments based on documents composed after the outcomes were known. For example, performance may drop when predicting judgments in years marked by significant legal or societal developments, as observed by Medvedeva et al. [9]

Moreover, the binary classification nature of the task simplifies the prediction process but may not fully capture the complexity of legal reasoning. Predicting violation or non-violation does not account for the nuanced arguments or specific legal articles cited in the final rulings.

Forecasting systems like JURI SAYS provide valuable tools for understanding legal trends, preparing legal arguments, or supporting judicial decision-making. However, their predictions should be interpreted cautiously, given the potential for performance degradation when dealing with future cases.

3 Choice of an algorithm

Legal information, while written in natural language, often follows a specific and formalized style. However, this information is typically unstructured, posing challenges for automated processing. To handle large volumes of legal data effectively, we must rely on techniques developed within the field of natural language processing (NLP).

The system I worked on is capable of automatically predicting the category (i.e., verdict) associated with a new case. To achieve this, we employ supervised learning. In this approach, the system is trained using textual data from numerous court cases, along with their corresponding judgments. During this "training phase", the system learns to identify patterns and features linked to each category of verdict (e.g., violation vs. no violation).

To assess the system's performance, it is then tested on cases it has not encountered before, where the judgment is withheld. During this "testing phase," the system analyzes the given input and predicts the most likely outcome. This process, known as classification, relies on the patterns and features the system identified as significant during training. The ultimate aim is to evaluate the system's ability to generalize and make accurate predictions on unseen data.

3.1 Support Vector Machines

The foundational work by Medvedeva et al. utilized Support Vector Machines (SVMs) as the baseline algorithm. SVMs are classic supervised learning models that excel at binary classification tasks, making them suitable for the "violation" versus "no violation" dichotomy in ECtHR cases. The SVM-based approach relied on n-grams, sequences of one or more words extracted from case texts, as features for classification.

Figure 1 provides an illustration of how the system works. The algorithm identifies the optimal hyperplane, which is essentially a line (or a surface in higher dimensions), that best separates data into distinct classes. In the illustration, this is represented by the central line dividing the stars and circles.

The support vectors are the data points closest to the hyperplane. The SVM algorithm aims to position the hyperplane such that the margin, or the distance between the hyperplane and the nearest data points on either side, is maximized. By achieving the largest possible margin, the model improves its ability to classify new, unseen data correctly.

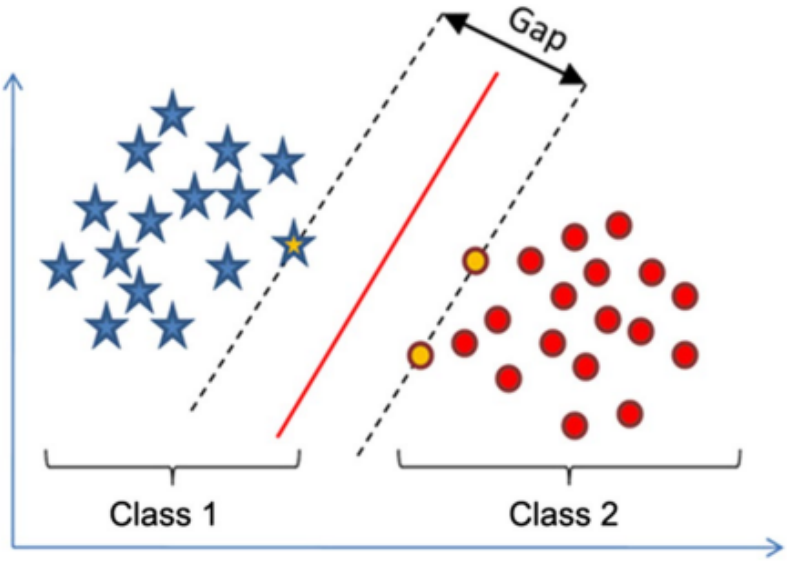

Figure 7 – Illustration of an SVM dividing data into classes. (Source : http ://diggdata.in/post/94066544971/support-vector-machine-without-tears)

Using communicated cases, this SVM model achieved an average accuracy of 75% across nine articles of the European Convention on Human Rights. This performance highlighted the model's potential for predicting case outcomes from early-stage data. However, Medvedeva et al. improved upon earlier work by Aletras et al. [2], addressing a critical flaw where court arguments referencing outcomes were included in training data. This correction ensured a more realistic evaluation of the model's predictive capabilities. While effective, the SVM-based approach faced limitations when applied to forecasting tasks. Its reliance on predefined features (n-grams) meant that it could not capture deeper semantic relationships or adapt to complex legal language variations, especially in early-stage documents.

4 Data

The research done by Medvedeva et al. relies on carefully curated datasets derived from the European Court of Human Rights' (ECtHR) publicly available documents.[4] The meticulous preparation and preprocessing of data ensure that models operate with relevant, high-quality information while addressing the complexities inherent in ECtHR cases.

The cases we used date back to 2017 or earlier. This is because we are comparing our results to actual outcomes, and given the typical length of judicial proceedings, by the time Medvedeva's original article was published in 2019, the most recent judgments available were for cases initiated in 2017 at the latest.

4.1 The ECtHR and the structure of its judgements

To better understand the data used in this study, it is essential to grasp the structure of the European Court of Human Rights (ECtHR) and the organization of its judicial documents.

The ECtHR, established in 1959, is an international court tasked with addressing individual and State applications alleging violations of rights guaranteed by the European Convention on Human Rights (ECHR). Applications are always filed against one or more State Parties to the Convention, but never against individuals.

The Court's composition reflects the number of States Parties to the ECHR. Each State provides one judge, who serves a non-renewable nine-year term. Cases are heard either by 7-member Chambers or, in exceptional circumstances, by the 17-member Grand Chamber. Importantly, the judge representing the State accused of a violation (the "national judge") always participates in hearings.

Before a case can be heard by a Chamber, it must first pass the admissibility stage, which is determined by a single judge (not the national judge). If deemed admissible, the case is then judged on its merits either by a Chamber within one of the Court's five Sections or, in special cases, by the Grand Chamber.

The Court's rulings are publicly available online and follow a relatively consistent structure :

- **Introduction :** Includes the case title (e.g., "Lawless v. Ireland"), date, Chamber, Section, and the composition of the Court (judges, president, registrar).
- **Procedure :** Details the procedural steps from the lodging of the application to the final judgement.
- **Facts :**
- **Circumstances :** Provides relevant background information on the applicant and the events leading to the alleged rights violation.
- **Relevant Law :** Lists relevant provisions from other legal documents, such as domestic laws, European treaties, and international agreements.
- **Law :** Outlines the Court's legal reasoning, addressing each alleged violation individually.
- **Judgement :** States the Court's decision for each alleged violation.
- **Dissenting / Concurring opinions :** Includes additional opinions from judges, explaining their agreement with the majority (concurring) or their disagreement (dissenting).

This structured format ensures clarity in the Court's decisions and facilitates analysis of its rulings for research purposes.

4.2 Binary classification of Court decisions

TThe ECtHR often addresses multiple complaints within a single case, even when they relate to the same article of the European Convention on Human Rights (ECHR). However, for this algorithm, the task is simplified as a binary classification problem : predicting either a "violation" or "non-violation" of an article.

To maintain consistency and clarity, cases involving multiple offenses under a single article, which could lead to both "violation" and "non-violation" rulings, are excluded. By focusing only on cases with a single outcome, the experiment aims to reveal specific patterns distinguishing violations from non-violations.

4.3 Balancing the dataset

For machine learning algorithms to perform effectively, the data used for training must be carefully curated. A sufficient quantity of data is essential, but ensuring a balanced representation of the target classes is equally critical. Without this, the algorithm might simply learn the distribution of the classes rather than identifying meaningful features.

For example, when training the algorithm to predict violations of Article 13, the dataset initially contained 1,230 "violation" cases and only 170 "non-violation" cases. This imbalance could lead the model to disproportionately predict "violation" outcomes, achieving high accuracy simply due to class prevalence. To counteract this, a balanced dataset was created by randomly selecting 170 "violation" cases to match the number of "non-violation" cases. The remaining "violation" cases were reserved for testing.

Figure 8 summarizes the final dataset used in the study. A balanced number of "violation" and "non-violation" cases were included for each article.

Article	'Violation' cases	'Non-violation' cases	Total	Test set
Article 2	57	57	114	398
Article 3	284	284	568	851
Article 5	150	150	300	1118
Article 6	458	458	916	4092
Article 8	229	229	458	496
Article 10	106	106	212	252
Article 11	32	32	64	89
Article 13	106	106	212	1060
Article 14	144	144	288	44

FIGURE 8 – Final number of cases per article [1]

4.4 Preprocessing

To prevent the algorithm from learning outcomes directly from textual cues, specific parts of the court decisions were excluded from the dataset. For instance :

- **Judgments :** The text explicitly stating the court's decision (e.g., "The Court has found a violation of Article 8") was removed. Allowing the algorithm to train on this information would be equivalent to feeding it the answer.
- **Law :** The detailed legal arguments and discussions of the judges were excluded, as this section is not accessible to parties prior to the trial.
- **Dissenting / Concurring Opinions :** These sections, which provide additional perspectives from judges, were also excluded.

The remaining text included the Procedure, Circumstances, and Relevant Law, as these sections contain information available before the final judgment is made.

5 Reproducing Medvedeva's research

5.1 Algorithm and type of prediction

After training an SVM, we evaluate the performance of the model using a separate set of cases (test set) that were not included during training. The program predicts whether each case constitutes a violation or not, and these predictions are compared to the actual decisions of the court. The performance of the system is then measured as the percentage of correctly classified decisions.

The measure of the performance involves **k-fold cross-validation** (in our case, cross validation is 10-fold). In this approach, the entire dataset available for training is divided into k parts. One part is withheld while the model is trained on the remaining *k-1* parts. After training, the model's performance is evaluated by testing it on the withheld portion. This process is repeated k times, each time withholding a different subset and training on the rest. For example, in a 5-fold cross-validation ($k = 5$), the dataset is split into five equal parts. In each iteration, one part (20% of the data) serves as the test set, while the remaining 80% is used for training. This process is repeated five times, ensuring that every part of the dataset is used both for training and testing.

Cross-validation helps identify the optimal parameters for the machine learning model and ensures its robustness by evaluating performance across different data subsets. This method increases the likelihood that the model will generalize well to unseen cases.

5.2 Features

One option to identify useful features is to manually craft features, such as specific types of issues raised in a case. Another approach involves automatically extracting features. For example, the model could consider all individual words or short sequences of consecutive words, allowing the machine learning algorithm to determine which words or sequences are most indicative of a violation or non-violation.

These sequences of words are formally known as **n-grams**, where a single word is a unigram, two consecutive words form a bigram, and three consecutive words are called trigrams. For instance, consider the following sentence :

By a decision of 4 March 2003 the Chamber declared this application admissible.

If we extract bigrams (two-word sequences), the features would include :

— *By a*, *a decision*, *decision of*, *of 4*, *4 March*, *March 2003*, *2003 the*, *the Chamber*, *Chamber declared*, *declared this*, *this application*, *application admissible.*

For trigrams (three-word sequences), the features would be :

— *By a decision*, *a decision of*, *decision of 4*, *of 4 March*, *4 March 2003*, *March 2003 the*, *2003 the Chamber*, *the Chamber declared*, *Chamber declared this*, *declared this application*, *this application admissible.*

Once features are extracted, their associated values for each case need to be determined. A simple approach is to assign binary values—1 if the n-gram appears in a case description and 0 if it does not. However, this method discards valuable information, such as how often an n-gram occurs in a document. Using frequency as a feature value is an improvement (e.g., *By a* : 100, *4 March* : 1, *never in* : 0). Yet, some words are naturally more common and occur frequently, regardless of their relevance to a case. For instance, the unigram *the* will appear more often than *application*, despite being less meaningful.

To address this, a common approach is to normalize n-gram frequencies by considering how many documents include each term. This method, called **term frequency-inverse document frequency (tf-idf)**, assigns higher weights to terms that are distinctive to a small number of cases and lower weights to common, less distinctive terms. The formula, as implemented in the scikit-learn Python library [10], is :

$$\text{tf-idf}(d,t) = \text{tf}(t) \times \text{idf}(d,t),$$

where

$$\text{idf}(d,t) = \log\left(\frac{n}{\text{df}(d,t)}\right) + 1.$$

Here, n is the total number of documents, and $\mathrm{df}(d, t)$ is the number of documents containing term t. For example, if a document contains 1,000 words and the term *torture* appears 3 times, the term frequency (tf) is $\frac{3}{1000} = 0.003$. If *torture* appears in 10 out of 10,000 documents, the inverse document frequency (idf) is $\log\left(\frac{10{,}000}{10}\right) + 1 = 4$. The resulting tf-idf score is :

$$\text{tf-idf} = 0.003 \times 4 = 0.012.$$

This score reflects the distinctiveness of *torture* across the dataset, giving it a higher weight than the raw frequency, as it rarely appears in other documents.

5.3 Results

The model achieved an average accuracy of 75% when predicting violations of nine articles of the ECHR. This result demonstrates the feasibility of using machine learning approaches in the legal domain, even with relatively simple models like SVMs. However, the performance varied significantly depending on the specific article being analyzed, largely due to differences in the availability of training data. Articles with fewer examples exhibited poorer classification performance, suggesting that data scarcity directly affects the model's ability to generalize.

Misclassifications provided additional insights into the limitations of the model :

- **Context-dependent errors :** Cases with conflicting signals (e.g., violation-related terms appearing in non-violation cases) were particularly prone to errors. These instances highlight the model's reliance on surface-level patterns rather than deeper contextual understanding.
- **Class imbalance :** Differences in the distribution of violation and non-violation cases across articles likely contributed to biased predictions. For example, if most cases in the training set for a specific article involved violations, the model might overpredict violations in the test set.

The results I obtained in this project are, unsurprisingly, not exactly identical to those reported by Medvedeva (see figure 0 and table 1 below). However, they remain remarkably similar overall. For instance, it is worth noting the recall values for Article 2, where the violation and non-violation categories appear to be swapped between the two sets of results. This discrepancy is likely due to the proximity of the two values (0.74 and 0.68), which are relatively close and could explain the apparent "inversion" as a mere effect of chance.

Interestingly, the number of cases available for cross-validation does not seem to have a significant impact on the results. For example, Article 11, which had the fewest cases (just 64), still yielded values very similar to those found in Medvedeva's study. Conversely, Article 10, which had a much larger dataset (212 cases), produced results that, while close to those obtained by Medvedeva, are somewhat surprising. Specifically, precision, recall, and F1-score for Article 10 are consistently 0.66 across all metrics. This observation suggests that the number of cases does not necessarily correlate with the accuracy or consistency of the results.

Further evidence supporting this hypothesis emerged when I built upon Medvedeva's experiment, which I will elaborate on in a subsequent section.

5.4 Performance

Art #	Class	Precision	Recall	F-score
Art 2	Non-violation	0.72	0.68	0.70
Art 2	Violation	0.70	0.74	0.72
Art 3	Non-violation	0.80	0.77	0.79
Art 3	Violation	0.78	0.81	0.80
Art 5	Non-violation	0.77	0.75	0.76
Art 5	Violation	0.76	0.77	0.77
Art 6	Non-violation	0.78	0.87	0.82
Art 6	Violation	0.85	0.76	0.80
Art 8	Non-violation	0.69	0.76	0.72
Art 8	Violation	0.73	0.66	0.69
Art 10	Non-violation	0.63	0.66	0.65
Art 10	Violation	0.64	0.61	0.63
Art 11	Non-violation	0.86	0.78	0.82
Art 11	Violation	0.80	0.88	0.8
Art 13	Non-violation	0.83	0.86	0.85
Art 13	Violation	0.85	0.83	0.84
Art 14	Non-violation	0.77	0.76	0.77
Art 14	Violation	0.77	0.77	0.77

FIGURE 9 – Precision, recall and f-score per class per article obtained by Medvedeva [1]

Article	Class	Precision	Recall	F-Score
Article 2	Non-Violation	0.70	0.75	0.73
	Violation	0.74	0.68	0.71
Article 3	Non-Violation	0.81	0.79	0.80
	Violation	0.79	0.81	0.80
Article 5	Non-Violation	0.75	0.73	0.74
	Violation	0.74	0.75	0.75
Article 6	Non-Violation	0.77	0.86	0.82
	Violation	0.85	0.75	0.79
Article 8	Non-Violation	0.69	0.75	0.72
	Violation	0.72	0.66	0.69
Article 10	Non-Violation	0.66	0.66	0.66
	Violation	0.66	0.66	0.66
Article 11	Non-Violation	0.86	0.78	0.82
	Violation	0.80	0.88	0.84
Article 13	Non-Violation	0.84	0.87	0.85
	Violation	0.86	0.83	0.85
Article 14	Non-Violation	0.75	0.78	0.77
	Violation	0.77	0.74	0.75

Tableau 1 – Precision, Recall, and F-Score per class perarticle obtained by me

The performance of the algorithm is evaluated using accuracy and the F1-score, two standard metrics in machine learning.

- **Accuracy :** Measures the proportion of correct predictions over all cases. It is straightforward but can be misleading if the dataset is imbalanced (e.g., more "violation" cases than "no violation" cases).
- **Precision :** The proportion of predicted "true" outcomes that are actually correct. It reflects how precise the model's predictions are.
- **Recall (Sensitivity) :** The proportion of actual "true" outcomes that the model correctly identifies. It captures the model's ability to detect all relevant cases.
- **F1-Score :** A harmonic mean of precision and recall, designed to balance the trade-off between the two. This metric is particularly useful when precision and recall need to be balanced, as improving one often comes at the expense of the other (e.g., in legal or medical applications).

While accuracy gives a broad picture, the F1-score is more informative in legal judgment forecasting, where the cost of false positives or false negatives can be high. By incorporating both precision and recall, the F1-score ensures a fair assessment of the model's performance in scenarios with imbalanced data or nuanced outcomes.

6 Building on Medvedeva's research

6.1 New experiment

The task I undertook based on Medvedeva's work serves two main purposes :

1. **Practical application :** The predictive aspect of this task has clear practical implications. It could potentially assist judges by streamlining the process of determining whether there is a violation of a specific article of the ECtHR. This predictive tool could save valuable time in judicial decision-making.
2. **Analytical insight :** Beyond its practical application, this task offers an opportunity to investigate the textual signals present in judicial decisions. Legal scholars often argue that judgments are normative texts, encapsulating the rationale for decisions and reflecting the underlying reasoning of the court. This raises interesting questions : To what extent do textual features reflect these judicial decisions ? Are there systematic linguistic patterns that align with cases of violation versus non-violation ? By analyzing these patterns, this task contributes to a deeper understanding of the drafting conventions in judicial decisions. Additionally, it provides insights into how textual signals may have evolved over time, particularly in light of significant changes in judicial practices or socio-cultural contexts.

6.2 Task description and methodology

The classification task I chose to undertake aimed at categorizing cases into distinct temporal categories to identify trends or differences over time. This classification was designed to account for potential influences of socio-cultural changes on the drafting of judicial decisions.

To achieve this, I built on the directory structure and methodology proposed by Medvedeva. Specifically, her code relied on an architecture of nested directories, with one folder for each article of the ECtHR (e.g., a folder for Article 2, another for Article 3, and so on). Each article-specific folder was further divided into two subfolders : `violation` and `non-violation`. These subfolders contained text files, each corresponding to a specific case. Since the code depended on this hierarchical directory structure, it was essential to replicate it in my implementation.

With the assistance of M. Tarissan, I developed a script to duplicate the original directory structure. Subsequently, I created a program to extract the year of the case from the text files. Initially, there were some issues with date recognition (e.g., the script incorrectly identified dates such as the year 3080). I resolved these issues by specifying the expected date format (e.g., day, month, and year, such as *4 September 2004*).

Once the year extraction process was reliable, I implemented a script to sort cases into the appropriate folders based on their dates. Initially, I categorized cases into two broad groups : those predating 2010 and those from 2010 or later. This categorization worked seamlessly.

On the advice of a researcher contacted by M. Tarissan, I later refined the categorization into three temporal categories : cases before 1980, cases between 1980 and 2011, and cases post-2011. The choice of 2011 was informed by significant structural changes, such as the introduction of the pilot judgment procedure into the rules of the European Court of Human Rights. Additionally, the Brighton Declaration (2012) criticized the court for being overly activist, suggesting that the years following 2011 could serve as an interesting reference point for a before-and-after comparison.

Initially, I attempted a division into the periods 1950–1980, 1980–2011, and 2011–2017. However, this approach proved unfeasible due to insufficient data for the earlier period (e.g., for Article 8, only five cases were available before 1980). Consequently, I opted for a more balanced division : cases before 2000, cases between 2000 and 2011, and cases post-2011. This adjustment ensured a sufficient number of cases in each temporal category for meaningful analysis.

6.3 Results

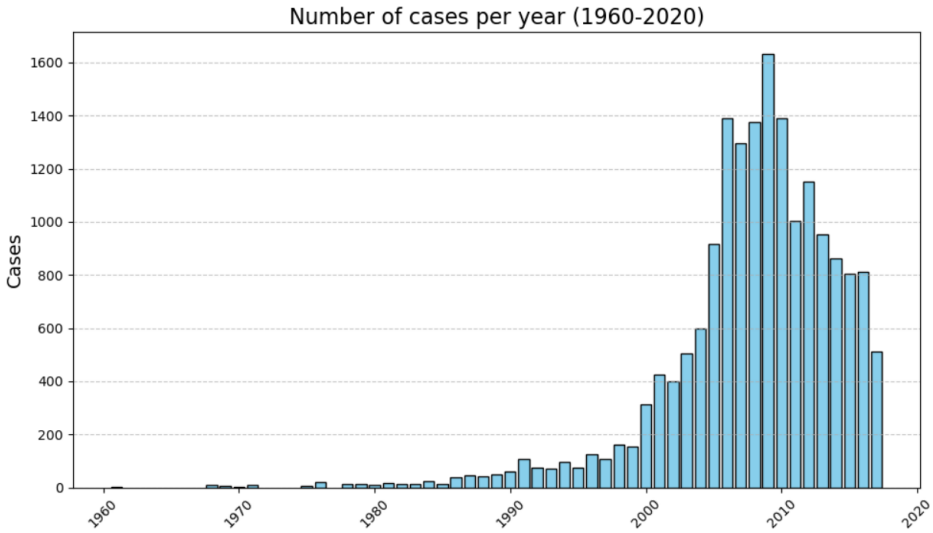

FIGURE 10 – Number of cases per year

All performance results and number of cases per period can be found in the annex.

The results obtained for Article 2 before 2010 were not particularly strong, which may be partly explained by the limited number of available cases. Specifically, only 47 cases were available for this period, compared to 114 cases across all time periods. In contrast, for Article 3, while the number of cases before 2010 was almost half that of the total dataset (237 cases before 2010 versus 668 overall), the performance metrics remained extremely similar. This suggests the hypothesis that there may exist a threshold above which the performance becomes comparable, even with fewer data points. However, this hypothesis should be approached with caution, as the dataset is not particularly large overall. For each article, there are at most a few hundred cases, which is far from the scale of tens of thousands of cases typically seen in larger datasets. Nonetheless, given the practical constraints of the legal domain (e.g., the finite number of cases available each year), working with a smaller dataset is still valuable and meaningful.

Aside from Article 2 and Article 11, where the number of cases is particularly low (47 and 39 cases, respectively), the performances obtained for the period before 2010 are very similar to those observed across the entire dataset.

For the period after 2010, the hypothesis regarding the impact of the number of cases does not hold consistently. While the results for Article 11 (with only 25 cases) are indeed very poor, this is not true for Article 13, which also has a low number of cases (43 cases) but does not exhibit the same drop in performance. In general, the results for this period are more variable than for the period before 2010, likely due to the smaller dataset available for this timeframe. This highlights the importance of attempting to balance the number of cases across time periods.

However, achieving such balance is challenging. While the key transitional years (2011 and 2012) have been identified, the vast majority of the available cases are concentrated between 2000 and 2011. Selecting too short a period would limit the representativeness of the data, while overly unbalanced periods (e.g., one very long and one very short) make meaningful comparisons difficult. Therefore, balancing the length of periods and the number of cases within them remains a significant challenge.

Another difficulty is the lack of post-2011 cases for certain articles, such as Articles 11, 13, and 14, in the dataset at our disposal.

One notable finding is the clear difference in performance between violation and non-violation cases. Overall, the performance is significantly lower for cases classified as violations of the relevant article, despite an equal number of violation and non-violation cases being used during training. This discrepancy

is surprising and warrants further investigation.

The results for the post-2011 period, when compared to the performance across the entire dataset, are mixed. For some articles, such as Article 5, the results are very similar, while for others, such as Article 10, there are substantial differences. These differences do not appear to be directly linked to the number of cases used for training. For example, Article 10 had 75 cases, while Article 5 had 97, a relatively small difference. Article 3, with the largest number of cases for the post-2011 period (291 cases), exhibits results that are relatively close to those obtained across the entire dataset, although the difference is still greater than that observed for Article 5. These findings suggest that factors other than the number of cases, such as the inherent characteristics of the articles or the textual features of the cases, may influence the performance.

7 Conclusion

7.1 Experience and key takeaways

The decision to begin my project by reproducing the results of Medvedeva's article was particularly important given my academic background. Starting directly with exploratory research would not have been a suitable approach for me at this stage ; I needed a solid reference framework to progress effectively. Reproducing the results provided a reliable benchmark in terms of performance metrics and allowed me to build confidence in the methodology. Successfully replicating the results with my own simulations has also enabled me to propose extensions to this work, particularly in a direction motivated by legal context—namely, focusing on the pivotal year 2011.

I greatly appreciated the balance between independent work and regular guidance throughout the project. I am deeply grateful to my supervisor, Mr. Tarissan, for his consistent support, not only during this project but also earlier in helping me secure my position in the ARIA program. This project was quite intense, but I learned an enormous amount during the process. While it was not always easy — particularly as I often progressed through trial and error — discussions with Mr. Tarissan and my peers revealed that this experience is a common aspect of working with code. For me, this project was as much a discovery of programming as it was an exploration of the field of artificial intelligence.

7.2 Future Directions

There are several avenues for further research that could extend the work I have undertaken. One possibility, previously discussed with Mr. Tarissan, would be to compare the performance of the Support Vector Machine (SVM) used in this project with that of another machine learning algorithm, such as Random Forest.

Another promising direction would involve working with a larger dataset. This could help address the data imbalance issues that I encountered in the later stages of the project, which limited the ability to fully evaluate some of the proposed methodologies.

Expanding the dataset and exploring alternative algorithms would not only enhance the robustness of the results but could also provide further insights into the textual features associated with legal judgments and their classification. These steps represent exciting opportunities to continue this line of research.

8 Annex

8.1 All years combined

Article	**Number of Cases**	
Article 2	114	
Article 3	568	
Article 5	300	
Article 6	916	
Article 8	458	
Article 10	212	
Article 11	64	
Article 13	212	
Article 14	288	

Tableau 2 – Number of Cases Per Article, all years

Article	Class	Precision	Recall	F-Score
Article 2	Non-Violation	0.70	0.75	0.73
	Violation	0.74	0.68	0.71
Article 3	Non-Violation	0.81	0.79	0.80
	Violation	0.79	0.81	0.80
Article 5	Non-Violation	0.75	0.73	0.74
	Violation	0.74	0.75	0.75
Article 6	Non-Violation	0.77	0.86	0.82
	Violation	0.85	0.75	0.79
Article 8	Non-Violation	0.69	0.75	0.72
	Violation	0.72	0.66	0.69
Article 10	Non-Violation	0.66	0.66	0.66
	Violation	0.66	0.66	0.66
Article 11	Non-Violation	0.86	0.78	0.82
	Violation	0.80	0.88	0.84
Article 13	Non-Violation	0.84	0.87	0.85
	Violation	0.86	0.83	0.85
Article 14	Non-Violation	0.75	0.78	0.77
	Violation	0.77	0.74	0.75

Tableau 3 – Precision, Recall, and F-Score by Class for Each Article

Article	Accuracy	Macro Avg	Weighted Avg	Confusion Matrix
Article 2	0.72	0.72	0.72	[[43, 14], [18, 39]]
Article 3	0.80	0.80	0.80	[[223, 61], [54, 230]]
Article 5	0.74	0.74	0.74	[[110, 40], [37, 113]]
Article 6	0.81	0.81	0.81	[[396, 62], [115, 343]]
Article 8	0.71	0.71	0.70	[[171, 58], [77, 152]]
Article 10	0.66	0.66	0.66	[[70, 36], [36, 70]]
Article 11	0.83	0.83	0.83	[[25, 7], [4, 28]]
Article 13	0.85	0.85	0.85	[[92, 14], [18, 88]]
Article 14	0.76	0.76	0.76	[[113, 31], [38, 106]]

Tableau 4 – Recapitulative Table with Accuracy, Averages, and Confusion Matrices

8.2 Key date 2010

8.2.1 2010 and before

Article	Number of Cases
2	47
3	237
5	180
6	685
8	249
10	124
11	39
13	169
14	202

Tableau 5 – Number of cases per article (cases selected from 2010 or earlier).

Article	Class	Precision	Recall	F1-Score
2	Non-Violation	1.00	0.11	0.19
2	Violation	0.62	1.00	0.77
3	Non-Violation	0.81	0.76	0.78
3	Violation	0.80	0.84	0.82
5	Non-Violation	0.78	0.76	0.77
5	Violation	0.76	0.78	0.77
6	Non-Violation	0.78	0.79	0.78
6	Violation	0.84	0.83	0.83
8	Non-Violation	0.65	0.72	0.68
8	Violation	0.69	0.62	0.65
10	Non-Violation	0.57	0.58	0.58
10	Violation	0.66	0.65	0.66
11	Non-Violation	0.59	1.00	0.74
11	Violation	0.00	0.00	0.00
13	Non-Violation	0.82	0.79	0.81
13	Violation	0.81	0.84	0.83
14	Non-Violation	0.77	0.85	0.81
14	Violation	0.81	0.72	0.76

Tableau 6 – Precision, Recall, and F1-Score for Non-Violation and Violation Classes.

8.2.2 2010 and after

Article	Number of Cases
Article2	67
Article3	331
Article5	120
Article6	231
Article8	209
Article10	88
Article11	25
Article13	43
Article14	86

Tableau 7 – Number of cases per article (2010 and after)

Article	**Class**	**Precision**	**Recall**	**F1-Score**
Article2	non-violation	0.59	1.00	0.75
Article2	violation	1.00	0.10	0.19
Article3	non-violation	0.80	0.75	0.77
Article3	violation	0.74	0.80	0.77
Article5	non-violation	0.78	0.78	0.78
Article5	violation	0.79	0.79	0.79
Article6	non-violation	0.72	0.98	0.83
Article6	violation	0.69	0.13	0.22
Article8	non-violation	0.73	0.79	0.76
Article8	violation	0.77	0.71	0.74
Article10	non-violation	0.65	0.82	0.72
Article10	violation	0.61	0.38	0.47
Article11	non-violation	0.00	0.00	0.00
Article11	violation	0.64	1.00	0.78
Article13	non-violation	0.80	0.96	0.87
Article13	violation	0.92	0.67	0.77
Article14	non-violation	0.67	0.54	0.60
Article14	violation	0.70	0.80	0.74

Tableau 8 – Precision, Recall, F1-Score by class for each article (2010 and after)

8.3 Key dates 2000 and 2011

8.3.1 2000 and before

Tableau 9 – Number of Cases Per Article before 2000

Article	Number of Cases
Article 3	31
Article 5	37
Article 6	131
Article 8	59
Article 10	39
Article 11	26
Article 13	26
Article 14	42

Tableau 10 – Precision, Recall, and F1-Score per Class before 2000

Article	Class	Precision	Recall	F1-Score
Article 3	Non-Violation	0.77	0.96	0.85
	Violation	0.00	0.00	0.00
Article 5	Non-Violation	0.70	1.00	0.83
	Violation	0.00	0.00	0.00
Article 6	Non-Violation	0.77	0.99	0.86
	Violation	0.86	0.17	0.29
Article 8	Non-Violation	0.67	0.97	0.79
	Violation	0.80	0.18	0.30
Article 10	Non-Violation	0.61	0.92	0.73
	Violation	0.33	0.07	0.11
Article 13	Non-Violation	0.88	1.00	0.94
	Violation	0.00	0.00	0.00
Article 14	Non-Violation	0.78	0.97	0.86
	Violation	0.83	0.38	0.53

8.3.2 2000 - 2011

Article	Number of Cases
2	48
3	246
5	166
6	595
8	220
10	98
11	31
13	155
14	174

Tableau 11 – Number of cases per article (2000 - 2011).

Article	Class	Precision	Recall	F-Score
2	Non-Violation	0.88	0.33	0.48
	Violation	0.65	0.96	0.78
3	Non-Violation	0.81	0.73	0.77
	Violation	0.81	0.87	0.84
5	Non-Violation	0.73	0.57	0.64
	Violation	0.70	0.82	0.76
6	Non-Violation	0.73	0.71	0.72
	Violation	0.83	0.84	0.83
8	Non-Violation	0.66	0.52	0.58
	Violation	0.67	0.79	0.72
10	Non-Violation	0.62	0.55	0.58
	Violation	0.73	0.78	0.76
11	Non-Violation	0.80	0.53	0.64
	Violation	0.67	0.88	0.76
13	Non-Violation	0.75	0.69	0.72
	Violation	0.79	0.84	0.81
14	Non-Violation	0.81	0.78	0.79
	Violation	0.79	0.82	0.80

Tableau 12 – Precision, Recall, and F-Score per class for each article (2000 - 2011).

8.3.3 2011 and after

Article	Number of Cases
Article 2	60
Article 3	291
Article 5	97
Article 6	190
Article 8	179
Article 10	75

Tableau 13 – Number of cases per article (2011 or later).

Article	Class	Precision	Recall	F1-Score
Article 2	Non-Violation	0.59	1.00	0.74
	Violation	1.00	0.04	0.08
Article 3	Non-Violation	0.78	0.73	0.76
	Violation	0.72	0.77	0.74
Article 5	Non-Violation	0.78	0.73	0.76
	Violation	0.75	0.79	0.77
Article 6	Non-Violation	0.74	0.99	0.84
	Violation	0.80	0.08	0.14
Article 8	Non-Violation	0.66	0.83	0.73
	Violation	0.73	0.52	0.61
Article 10	Non-Violation	0.62	0.82	0.71
	Violation	0.53	0.29	0.37

Tableau 14 – Precision, Recall, and F1-Score for Non-Violation and Violation per article. (2011 or after)

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